Routine Cardiac MRI: Systematic Approach to Interpretation

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See the slide presentation here.

Since its inception in the early 1990s, cardiac MRI (CMR) has evolved into a comprehensive method for diagnostic evaluation and prognostication of a wide variety of cardiac diseases affecting cardiac structure, hemodynamics, and function. The approach to diagnosis and interpretation can be overwhelming to a trainee attempting to navigate the multiple sequences and protocols. CMR studies may be tailored to address a specific question. Accordingly, the study may be modified to evaluate any unanticipated findings that are identified during imaging. Nonetheless, standard CMR sequences are those usually present, independent of the indication, which are the focus of the accompanying slide presentation.

The imaging study may begin with a set of preliminary chest MR images, which are useful for general assessment of the heart, major vasculature, and extracardiac structures. Dedicated CMR images are then obtained for the evaluation of anatomy, function, tissue characteristics, and hemodynamics. For each of these categories, information is provided in the slide presentation regarding the specific sequences to analyze, combine, and correlate for the purposes of diagnosis. The various types of chest MR images that may be obtained for quick assessment of the heart and vasculature are discussed, as well as potential important extracardiac findings.

In the anatomy section, we provide information on the proper cardiac plane definitions and nomenclature, emphasizing the normal appearance of each cardiac chamber, the cardiac valves, and the pericardium. The standard sequences utilized for this purpose include bright-blood imaging in the form of steady-state free-precession (SSFP) or spoiled gradient-echo (SGRE) sequences, both with cardiac gating. The SSFP sequence provides better signal-to-noise ratio (SNR) and has a fast acquisition time, while the SGRE sequence has a lower SNR and slower acquisition time but less susceptibility artifact and a higher sensitivity to flow. In the section of the slide presentation on function, which is evaluated with bright-blood cine images, we describe the lexicon used for ventricular, interatrial, and interventricular wall motion abnormalities and provide a brief overview of myocardial tagging.

In the tissue characterization section, we focus on the T1 and T2 characteristics of the myocardium and the behavior during early and late gadolinium interrogation. This provides information on the presence of edema, acute inflammation, and infiltration including iron deposition or fibrosis. Early and late gadolinium enhancement patterns can help suggest specific causes, and the extent of enhancement predicts the prognosis. Subtle changes in myocardial signal intensity characteristics can be difficult to visualize on conventional images. To overcome this obstacle, we describe the relatively recently introduced parametric imaging, which allows pixel-by-pixel quantification of T1, T2, and T2* values using short breath-hold mapping sequences. T1 and T2 signal intensity quantification by way of parametric map assessment and a brief overview of quantitative T2* imaging are included.

The hemodynamics section reviews myocardial perfusion, describes the mechanisms of flow quantification, and briefly addresses the valves in the setting of a nondedicated standard CMR study.

Understanding the normal CMR appearance is an essential element to help discern physiologic from pathologic features. The goal of the slide presentation is to ensure that the trainee can understand and recognize the normal CMR appearance of the heart, expected findings, and potential pitfalls as a first step in identifying an abnormality, rather than to provide a thorough review of specific diseases. For an optimal learning experience, a preexisting basic knowledge of CMR physics and dedicated cardiac sequences is desirable.

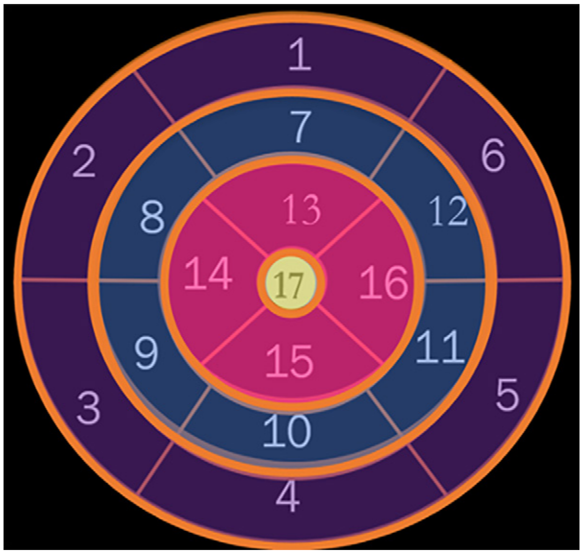

Figure 1. Polar map shows the short-axis distribution of regions of the heart. CMR mirrors the established echocardiogram myocardial nomenclature, which contains 17 segments (1–6 = base, 7–12 = midcavity, 13–16 = apical cavity, 17 = apex).

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Abbreviations: CMR = cardiac MRI, SGRE = spoiled gradient echo, SNR = signal-to-noise ratio, SSFP = steady-state free precession

TEACHING POINTS

- CMR studies should be evaluated according to four basic clinical questions: structural and anatomic anomalies, contractility disorders, myocardial tissue disease, and flow anomalies.
- The first step for accurate CMR interpretation is to recognize the normal anatomy and behavior of the heart on CMR images.
- Addressing each category individually and then as a group is the best way to gather all information necessary to provide a good differential diagnosis in each case.

After viewing the slide presentation, the trainee will be able to identify and recognize standard CMR sequences, nomenclature (Fig 1), normal cardiac chamber anatomy (Fig 2), normal tissue characteristics, use of parametric evaluation, and principles of flow hemodynamics as a springboard for more advanced learning.

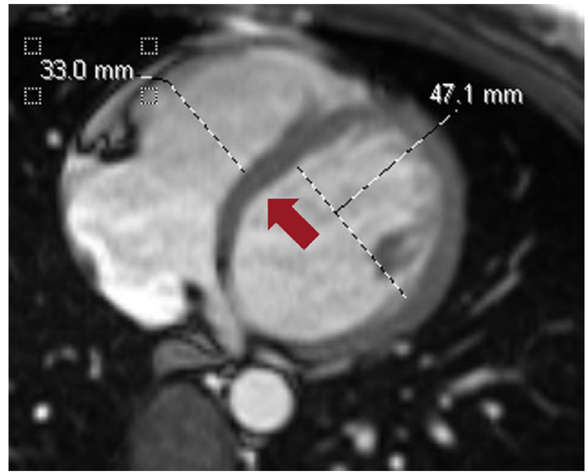

Figure 2. CMR image shows normal right and left ventricular sizes and normal positioning of the interventricular septum (arrow).

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Suggested Readings

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